

## Work Experience

### Junior System Reliability Engineer

Full Time

FMX

Boston, MA

May 2025 - Present

- Own the end-to-end development and deployment of an enterprise-grade web application providing the primary interface for our custom platform. Deployed and used daily for critical business needs across 10 environments.
- Implemented low-latency monitoring tooling using C and eBPF, enabling business-critical observability into trading system health.
- Automated the provisioning and deployment of 750 servers and 4000 software instances across 10 environments. Enabled and executed the repeatable recreation of an entire environment in 2 hours.

### Software Developer

Internship

FMX

New York, New York

June 2024 - May 2025

- Developed a network diagram automation tool that dynamically generated physical and logical network diagrams for over 200 office locations in 45 minutes. Leveraged Python concurrency and Jinja templating.
- Migrated all custom Kubernetes operators from Python 3.9 to Python 3.14, including the migration from multiprocessing to multithreading, reducing compute load by 30 percent.

### Co-Director

Internship

Plattsburgh State University Center for Cybersecurity and Technology

Plattsburgh NY

Jan 2024 - May 2024

- Elected as the student co-director for the largest technology organization at SUNY Plattsburgh due to academic performance, practical internship experience, and contributions to the organization.
- Hosted 1 workshop per week on various pragmatic applications of modern technologies. Some notable examples were on source control (git), Linux, and containerization.
- Responsible for managing three server racks owned by the organization, managing the secure implementation of firewalls, switching, remote access, and system administration.

## Projects

### Honeynet Distributed System

- Designed and implemented a distributed cybersecurity research platform that collected and analyzed real-world attack data using custom SSH/HTTP honeypots, centralized logging, and real-time packet inspection.
- Generated actionable threat intelligence from real-world attack telemetry, identifying attacker geolocations, commonly attempted username/password combinations, and repeated attempts to deploy a known botnet through the SSH honeypot.
- Led a 3-person engineering team and presented project findings to cybersecurity professionals at the **6th Annual North Country Cybersecurity Conference** after the project was selected to demonstrate the business value of honeypot-driven threat intelligence.

### Personal Website on AWS

- Deployed a personal portfolio website on AWS EC2, ensuring high availability and scalability. Implemented continuous deployment and integration via GitHub Actions and Docker Compose.
- Configured Nginx as a reverse proxy and Cloudflare for DNS to handle incoming traffic efficiently.

## Technologies and Languages

- **Languages:** Python, JavaScript, HTML, CSS, Java, SQL, C
- **Technologies:** SQLite, Postgres, AWS (EC2, S3), Docker, Kubernetes, Linux, Flask, Node JS, Git, ELK, TypeScript, REST

## Education: Plattsburgh State University

- **BSc:** Management Information Systems | **Minor:** Computer Science

Graduated May 2025